

INGENIERÍA DE SOFTWARE II

Práctica de laboratorio: Inspección de código con PMD

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Repositorio: <https://github.com/alejandrotapia20/CodeInspection>

1. Entorno

Eclipse 2026-09 con M2E, plugin PMD 7.28.0, Maven 3.9.16 y maven-pmd-plugin 3.21.2 (PMD 6.55.0). Usé JDK 26, compilando con release 21.

2. Maven en Eclipse

M2E ya venía incluido en Eclipse.

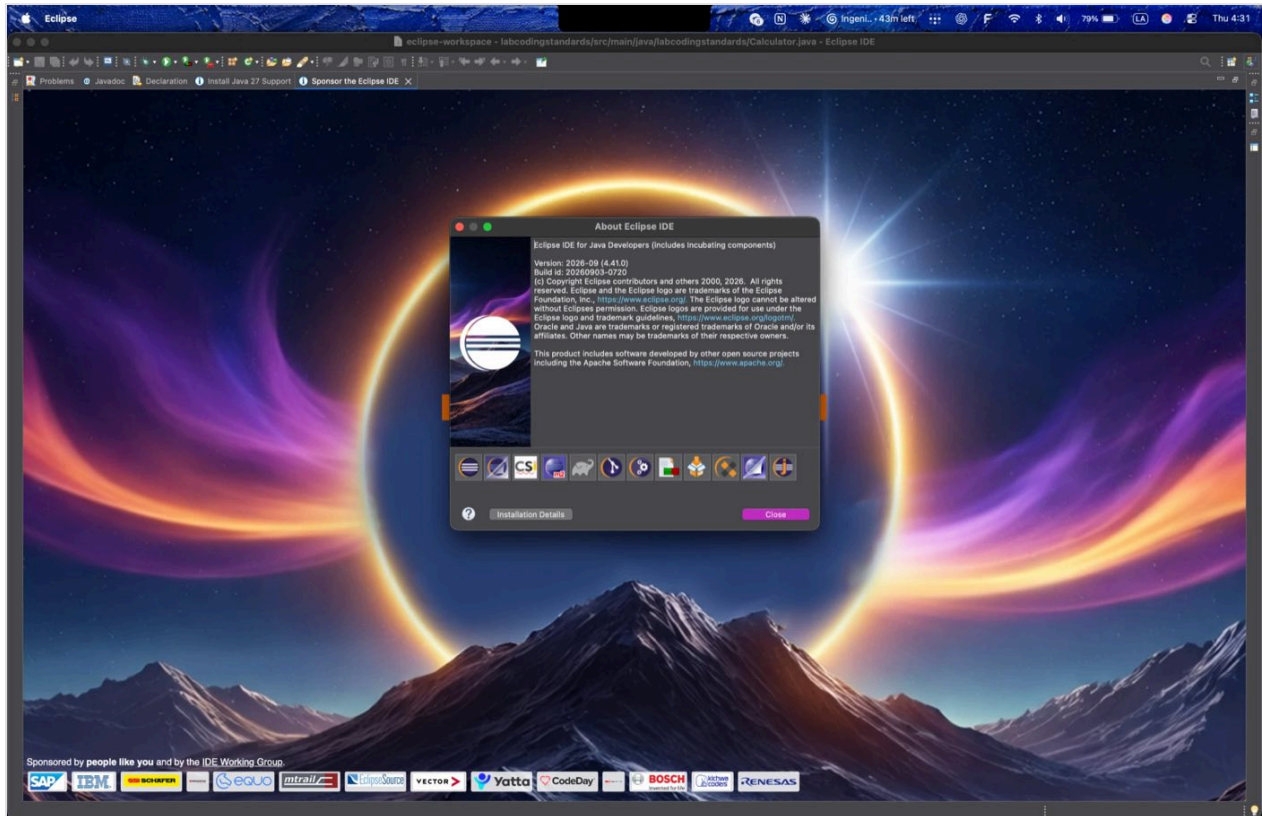


Figura 1. Versión de Eclipse.

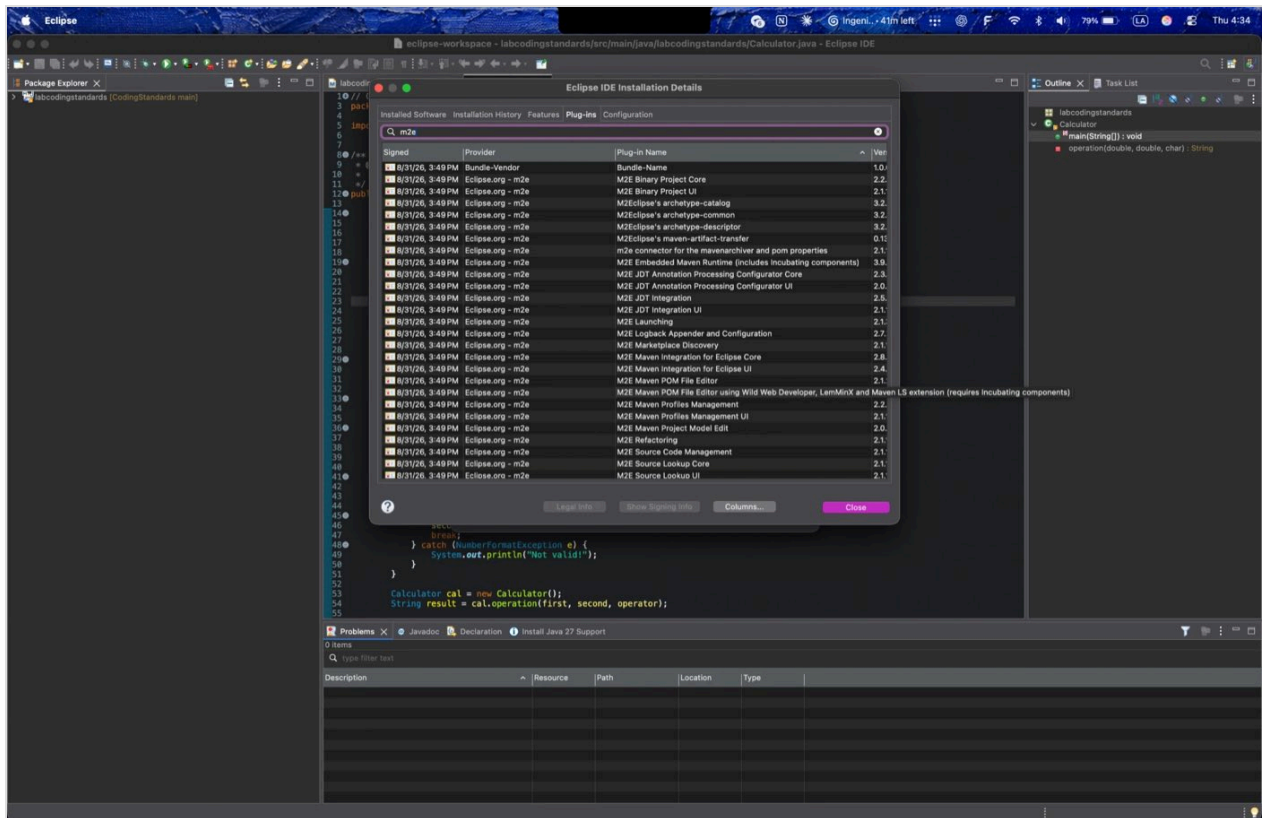


Figura 2. Plug-ins de M2E instalados.

3. Plugin PMD

Lo instalé desde <https://pmd.github.io/pmd-eclipse-plugin-p2-site/>.

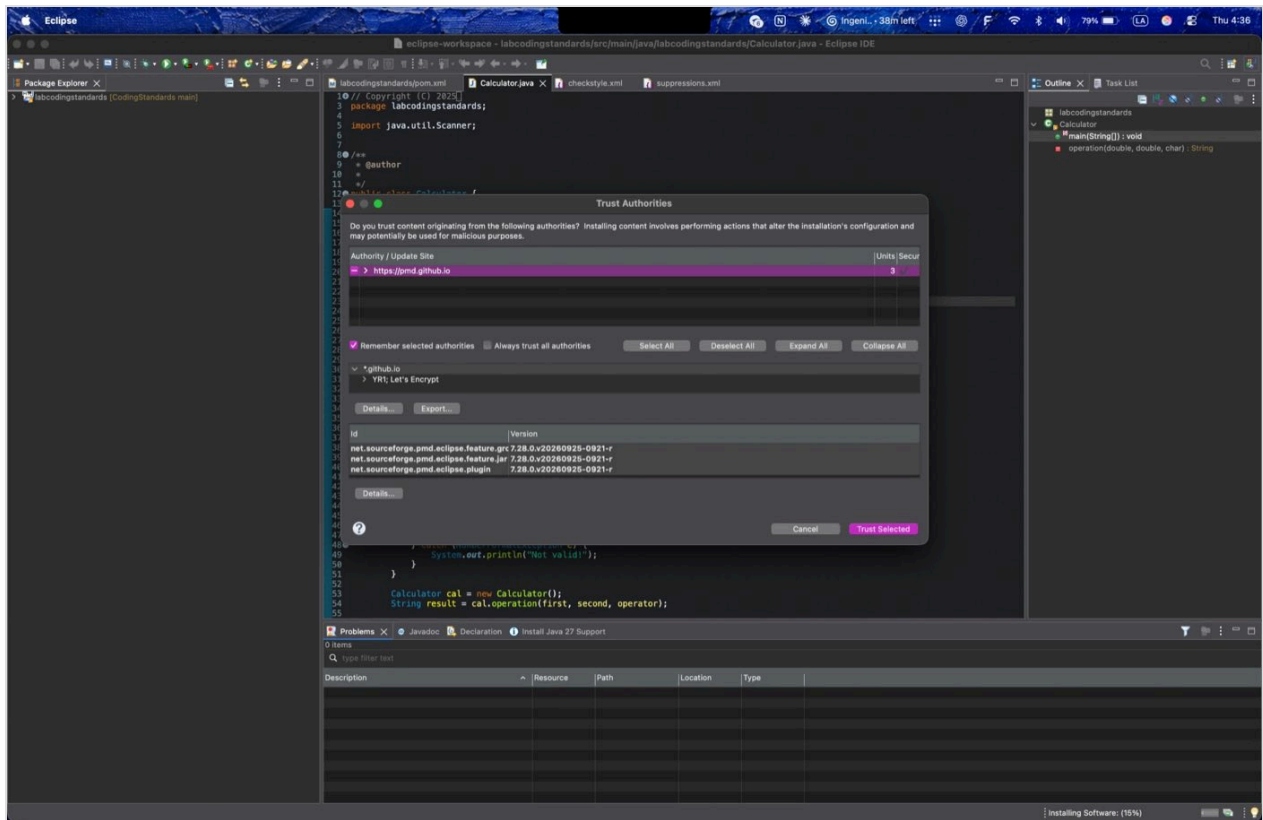


Figura 3. Instalación del plugin PMD 7.28.0.

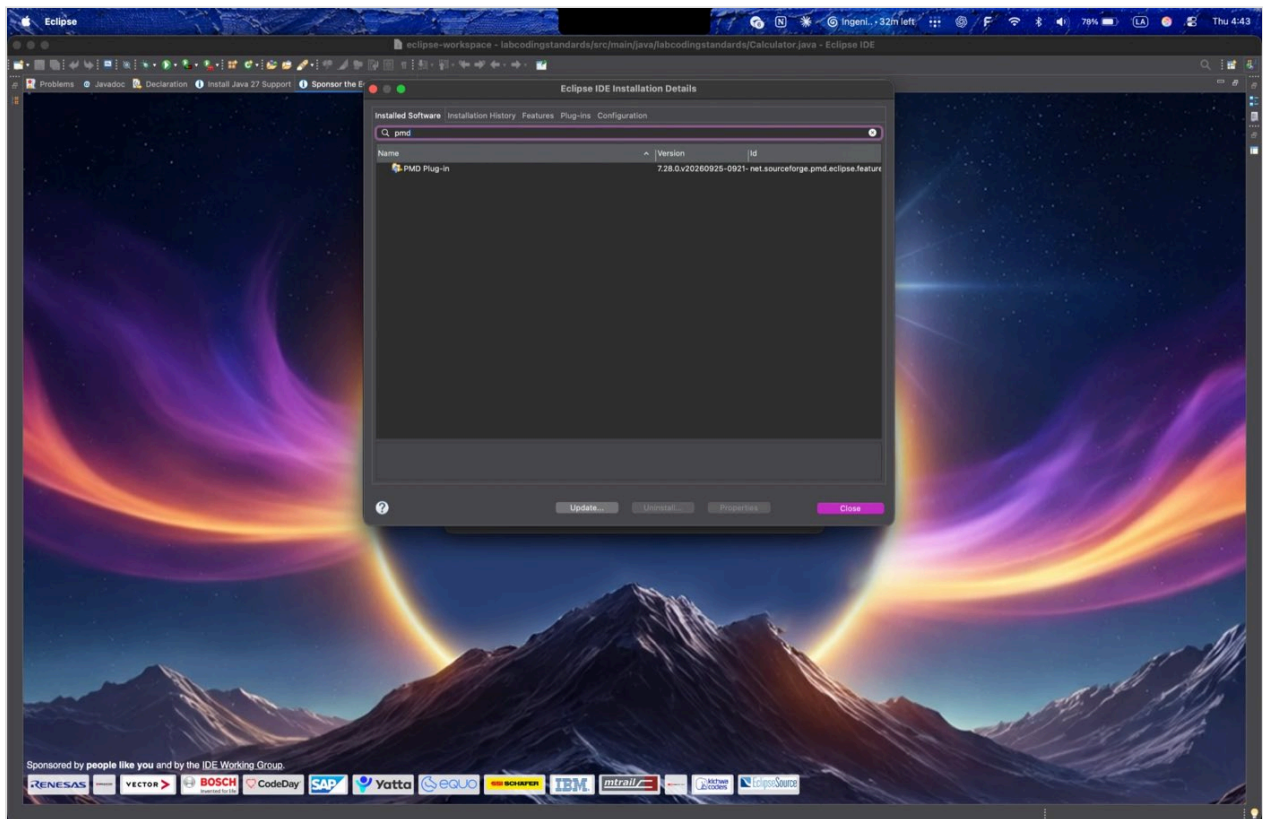


Figura 4. PMD Plug-in instalado.

4. Proyecto

Hice fork de leortyz/CodeInspection, lo cloné y lo importé como proyecto Maven. Como venía configurado para Java 1.5, le agregué `maven.compiler.release 21` al `pom.xml`. Activé PMD en el proyecto usando mi propio ruleset.

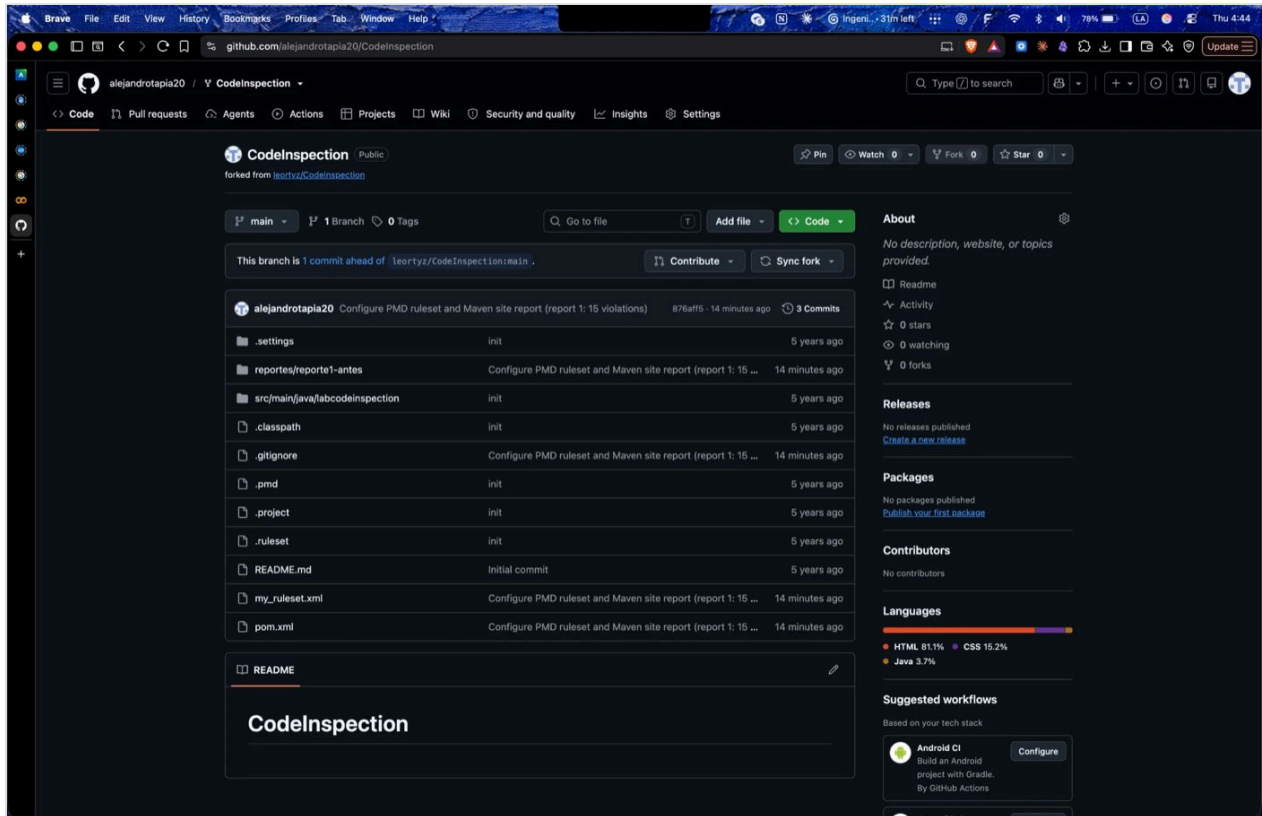


Figura 5. Fork en GitHub.

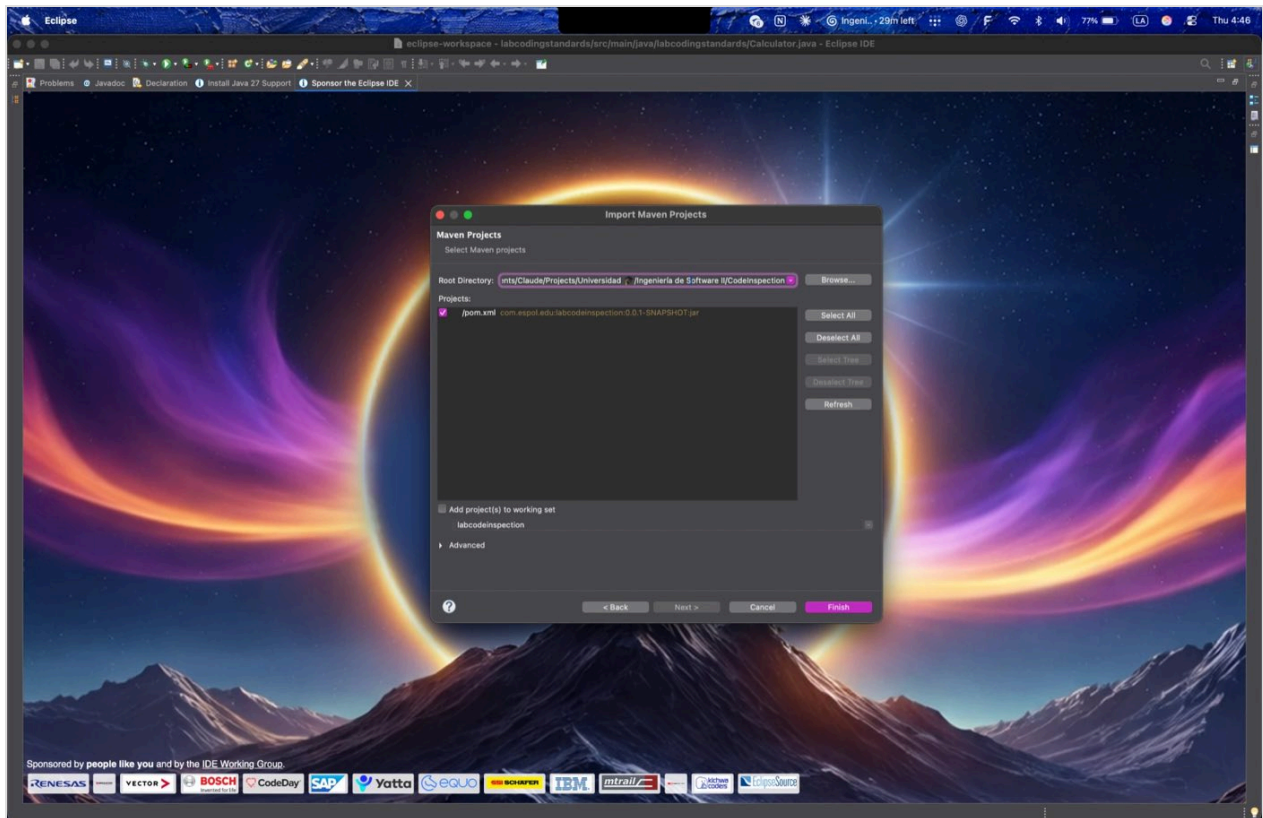


Figura 6. Importación del proyecto Maven.

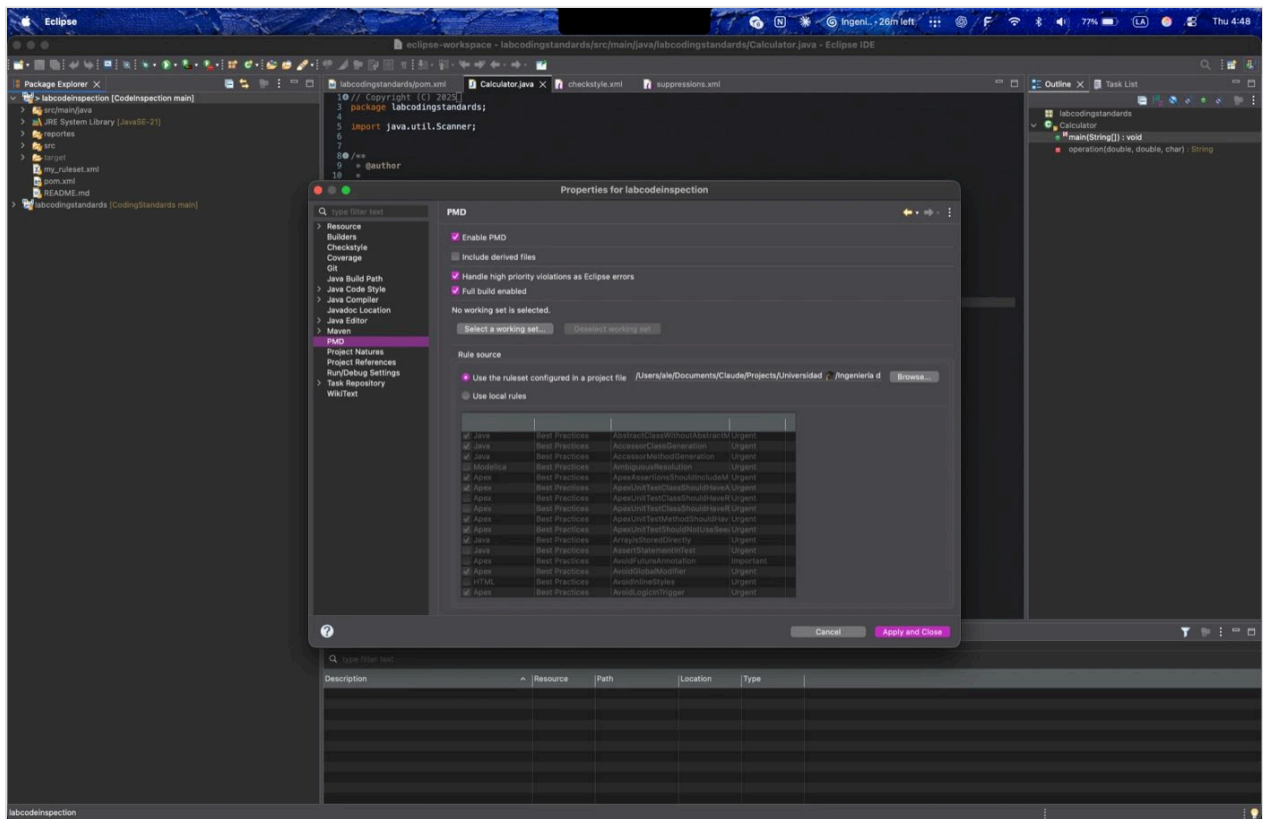


Figura 7. PMD habilitado con my_ruleset.xml.

5. Uso de PMD en Eclipse

Activé Check code after saving y corrí PMD → Check Code. Salieron 14 violaciones: 10 en Email.java y 4 en EmailApp.java.

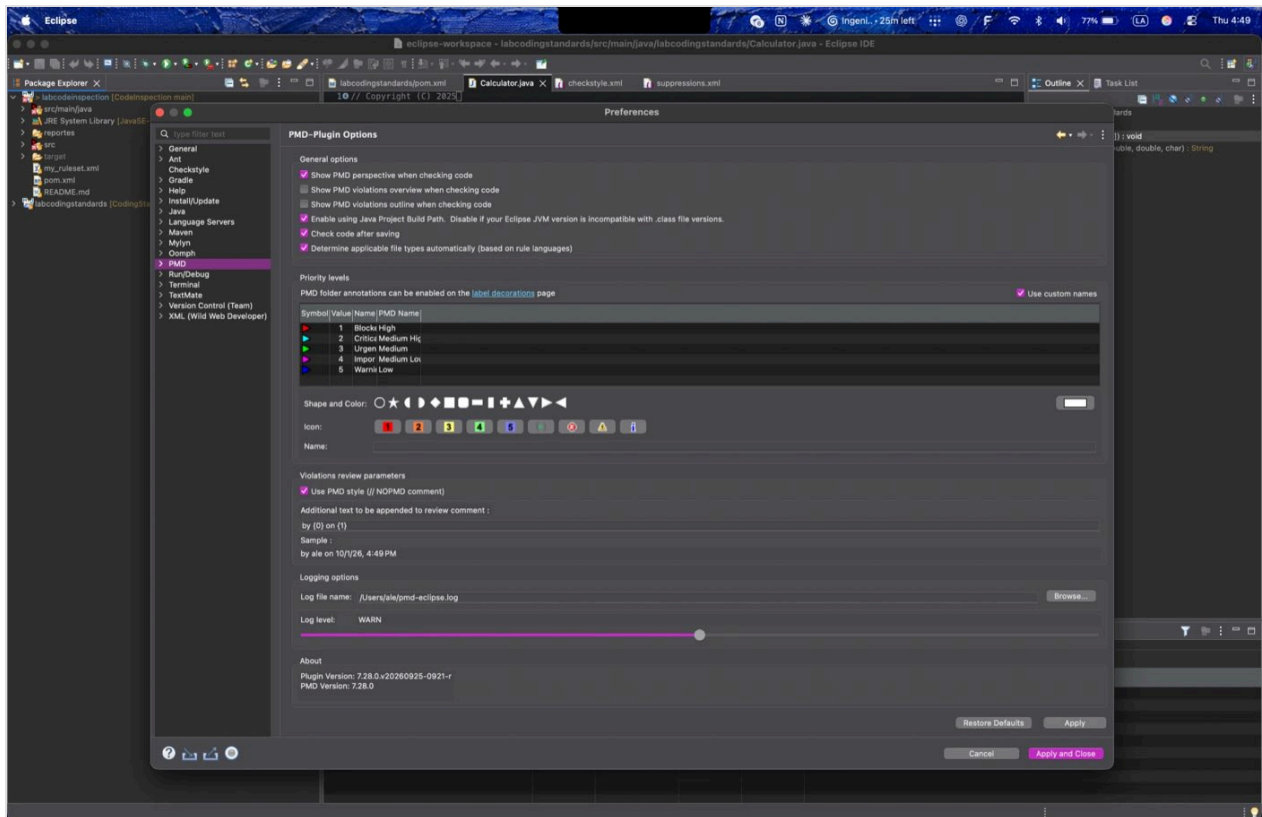


Figura 8. Check code after saving activado.

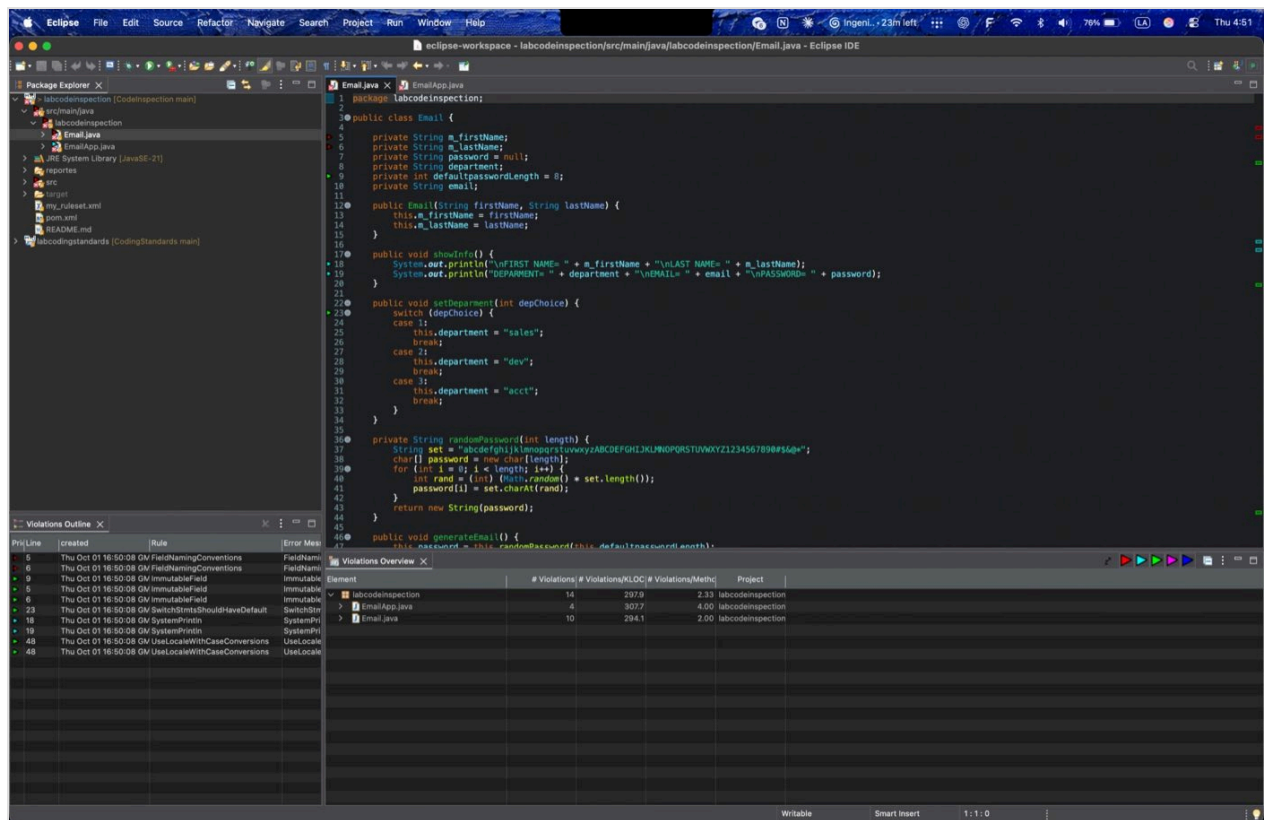


Figura 9. Violaciones en Email.java.

6. Conjunto de reglas

Armé my_ruleset.xml con 7 reglas que aplican al código del laboratorio: SystemPrintln, SwitchStmtsShouldHaveDefault, FieldNamingConventions, ShortVariable, ImmutableField, UseUtilityClass y UseLocaleWithCaseConversions.

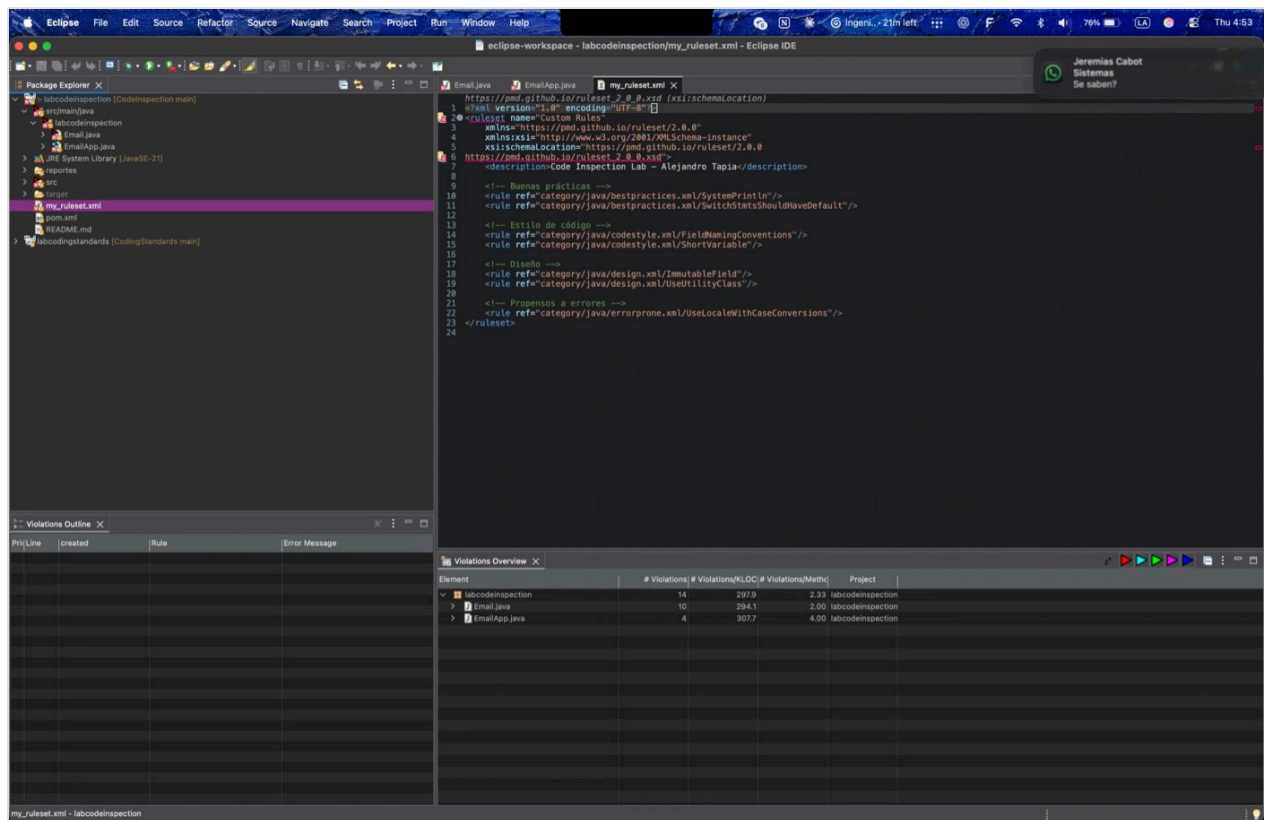


Figura 10. my_ruleset.xml.

7. Reportes con Maven

Agregué al `pom.xml` los plugins de la guía. Le puse `targetJdk 17` porque PMD 6.55 no soporta Java 21.

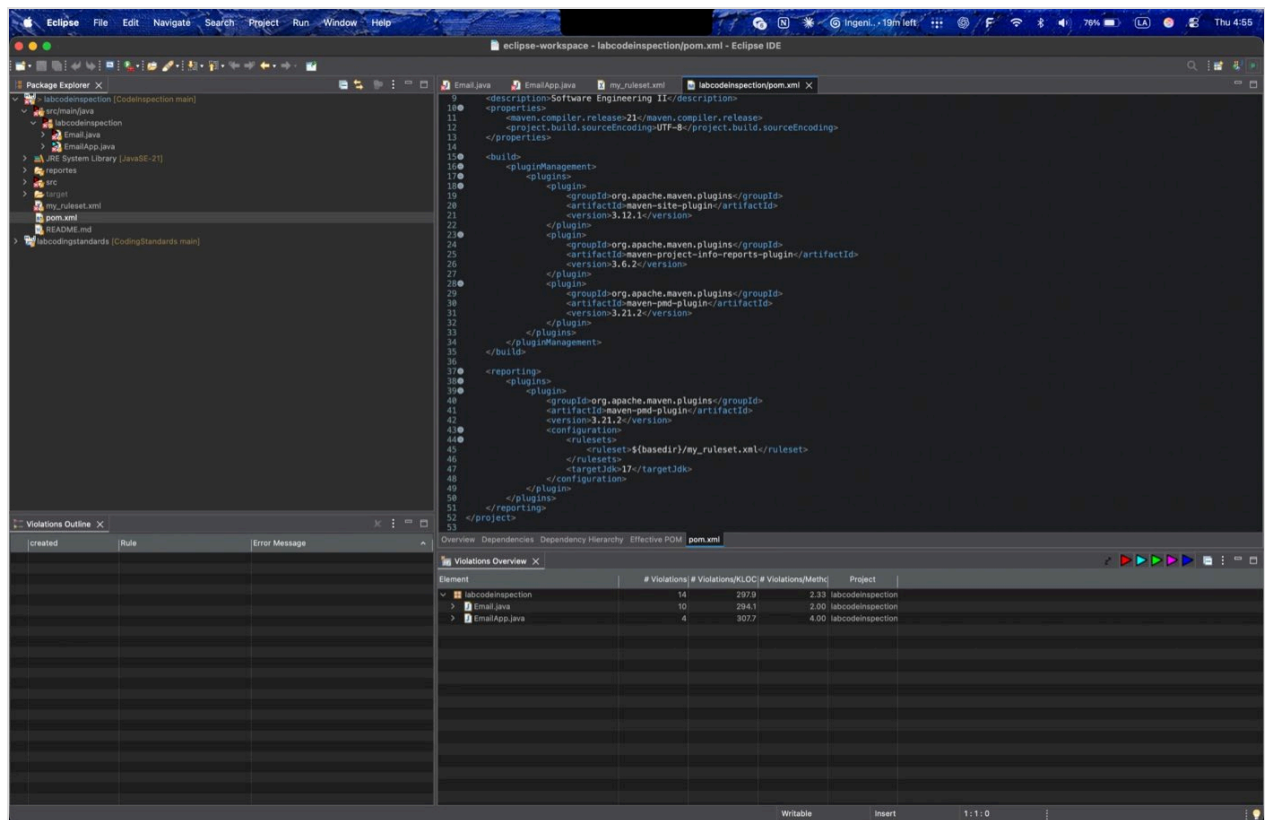


Figura 11. Sección reporting del pom.xml.

Reporte 1

mvn clean site reportó 15 violaciones. Maven da una más que Eclipse en EmailApp.java porque usa PMD 6 y Eclipse PMD 7.

Regla	Violaciones
SystemPrintln	5
ImmutableField	3
FieldNamingConventions	2
UseLocaleWithCaseConversions	2
SwitchStmtsShouldHaveDefault	1
UseUtilityClass	1
ShortVariable	1

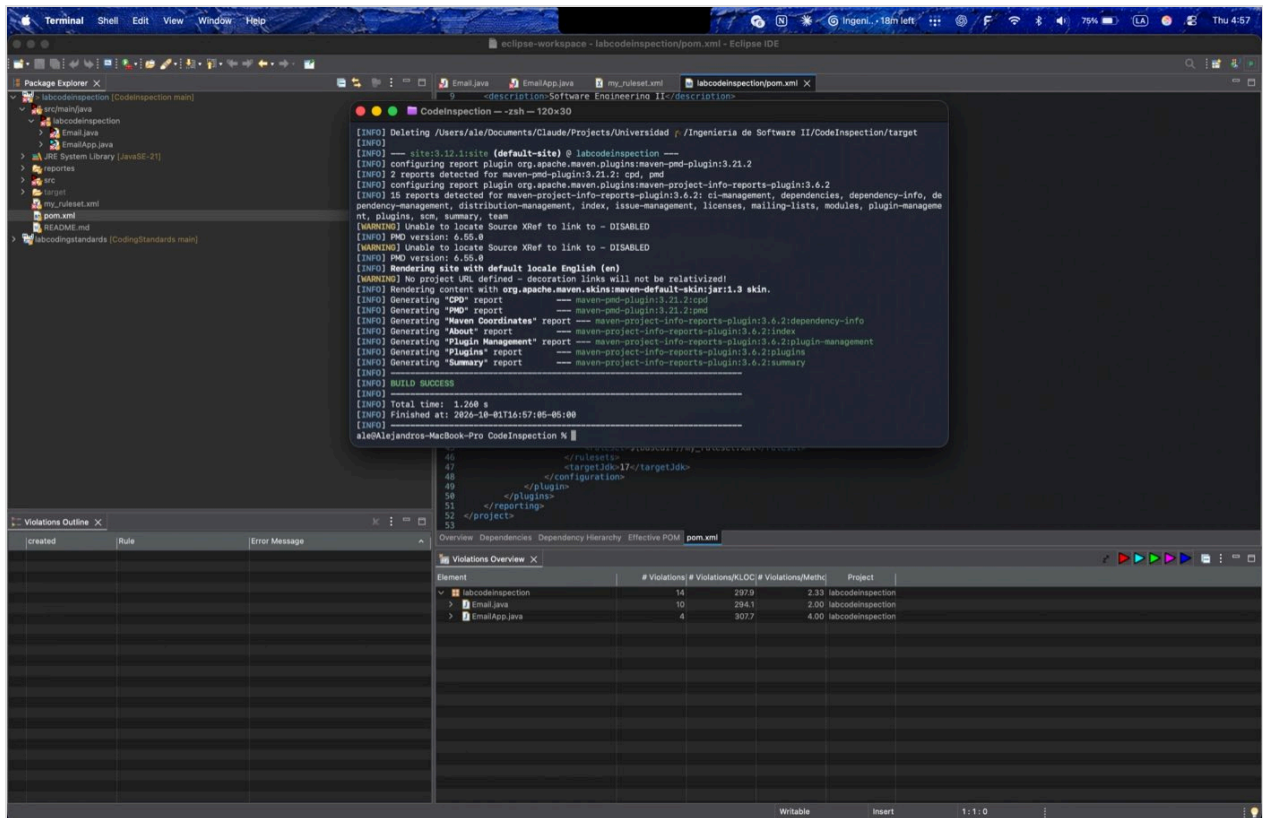


Figura 12. mvn clean site.

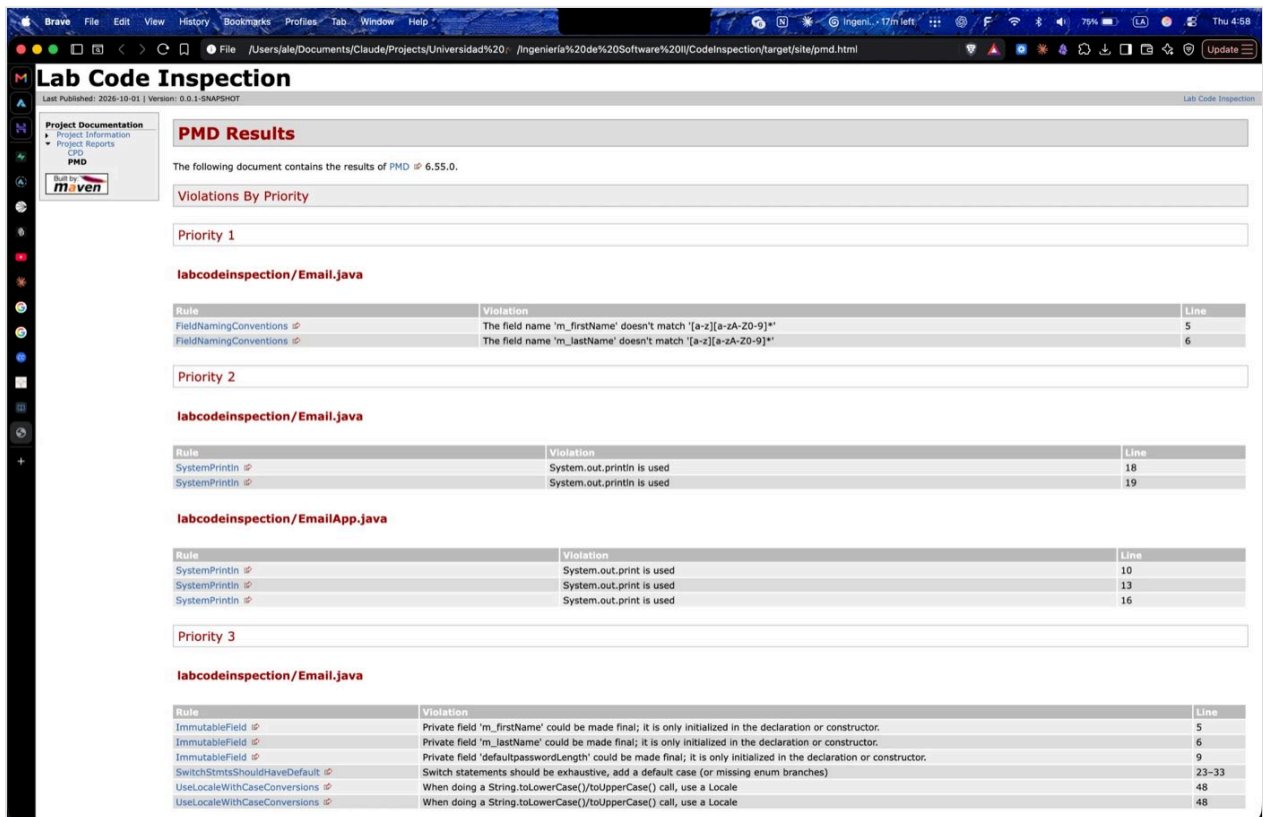


Figura 13. Reporte 1 por prioridad.

SystemPrintln	System.out.print is used	13	
SystemPrintln	System.out.print is used	16	

Priority 3

labcodeinspection/Email.java

Rule	Violation	Line
ImmutableField	Private field 'm_firstName' could be made final; it is only initialized in the declaration or constructor.	5
ImmutableField	Private field 'm_lastName' could be made final; it is only initialized in the declaration or constructor.	6
ImmutableField	Private field 'defaultpassword.length' could be made final; it is only initialized in the declaration or constructor.	9
SwitchStmtsShouldHaveDefault	Switch statements should be exhaustive, add a default case (or missing enum branches)	23-33
UseLocaleWithCaseConversions	When doing a String.toLowerCase()/toUpperCase() call, use a Locale	48
UseLocaleWithCaseConversions	When doing a String.toLowerCase()/toUpperCase() call, use a Locale	48

labcodeinspection/EmailApp.java

Rule	Violation	Line
UseUtilityClass	All methods are static. Consider using a utility class instead. Alternatively, you could add a private constructor or make the class abstract to silence this warning.	5-26
ShortVariable	Avoid variables with short names like sc	8

Files

labcodeinspection/Email.java

Rule	Violation	Priority	Line
ImmutableField	Private field 'm_firstName' could be made final; it is only initialized in the declaration or constructor.	3	5
FieldNamingConventions	The field name 'm_firstName' doesn't match '[a-z][a-zA-Z0-9]*'	1	5
ImmutableField	Private field 'm_lastName' could be made final; it is only initialized in the declaration or constructor.	3	6
FieldNamingConventions	The field name 'm_lastName' doesn't match '[a-z][a-zA-Z0-9]*'	1	6
ImmutableField	Private field 'defaultpassword.length' could be made final; it is only initialized in the declaration or constructor.	3	9
SystemPrintln	System.out.println is used	2	18
SystemPrintln	System.out.println is used	2	19
SwitchStmtsShouldHaveDefault	Switch statements should be exhaustive, add a default case (or missing enum branches)	3	23-33
UseLocaleWithCaseConversions	When doing a String.toLowerCase()/toUpperCase() call, use a Locale	3	48
UseLocaleWithCaseConversions	When doing a String.toLowerCase()/toUpperCase() call, use a Locale	3	48

labcodeinspection/EmailApp.java

Rule	Violation	Priority	Line
UseUtilityClass	All methods are static. Consider using a utility class instead. Alternatively, you could add a private constructor or make the class abstract to silence this warning.	3	5-26
ShortVariable	Avoid variables with short names like sc	3	8
SystemPrintln	System.out.print is used	2	10
SystemPrintln	System.out.print is used	2	13
SystemPrintln	System.out.print is used	2	16

Figura 14. Reporte 1 por archivo.

Correcciones

Cambié `m_firstName` y `m_lastName` a camelCase y los hice final junto con `defaultPasswordLength`. Le agregué default al switch, usé `Locale.ROOT` en `toLowerCase`, le puse constructor privado a `EmailApp` y renombré `sc` a `scanner`. Los `System.out` los dejé porque es una app de consola, y los suprimí con `@SuppressWarnings("PMD.SystemPrintln")`.

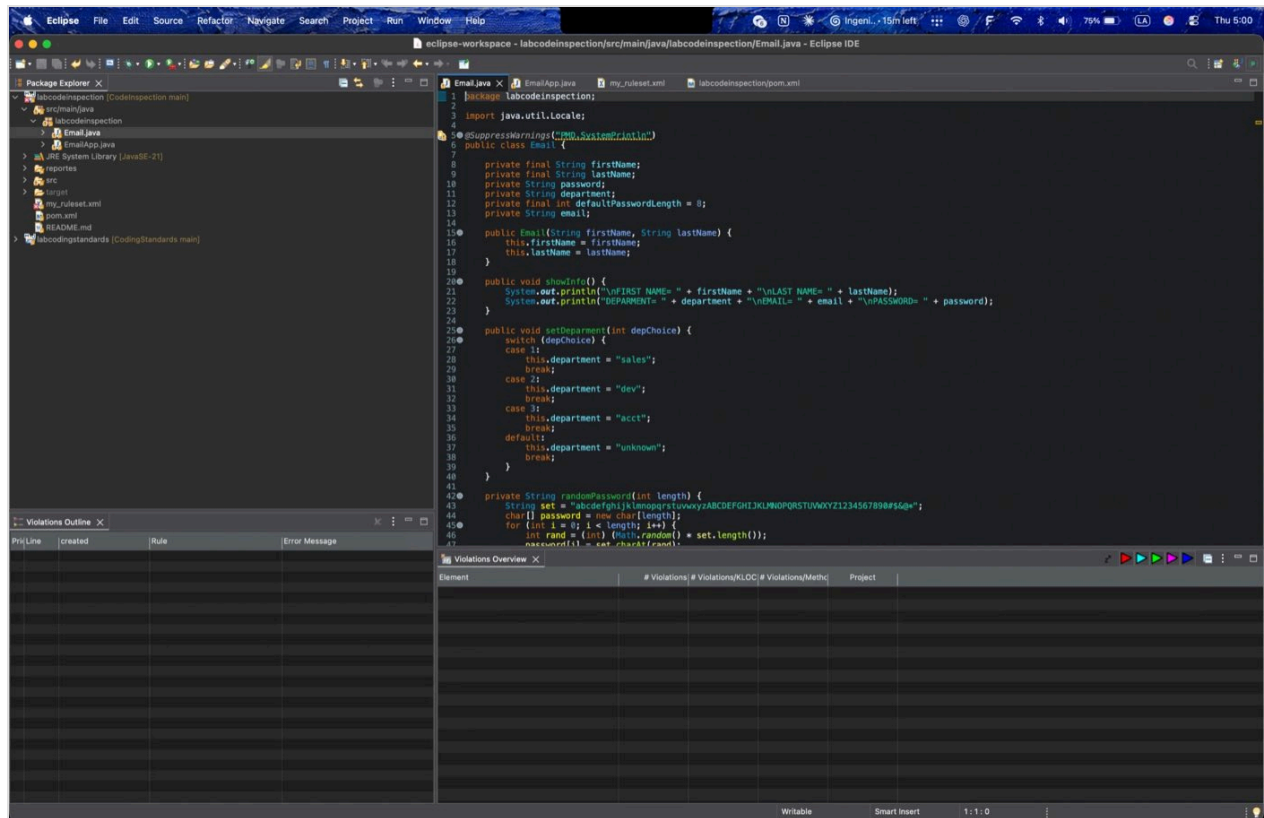


Figura 15. Email.java corregido, sin violaciones en Eclipse.

Reporte 2

Volví a correr mvn clean site y PMD ya no encontró problemas. Las salidas por consola aparecen como suprimidas.

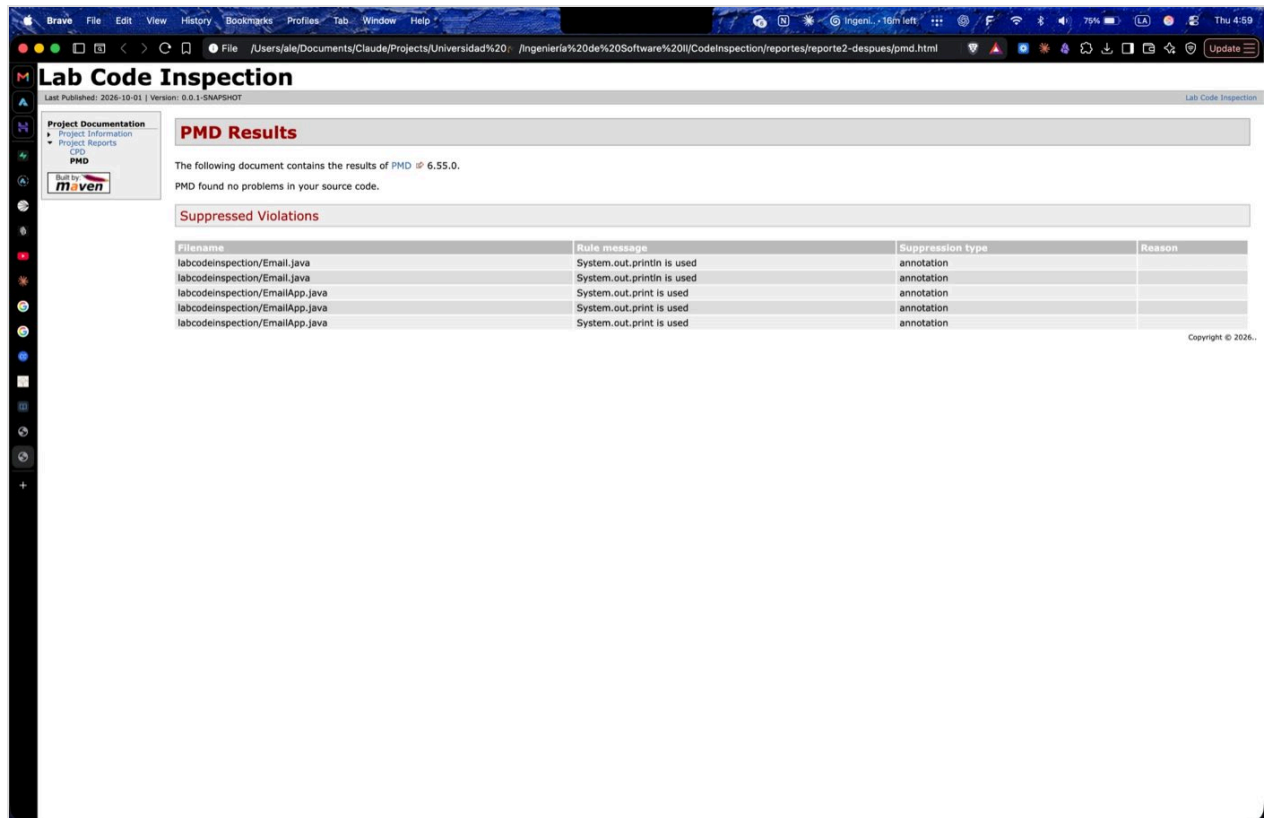


Figura 16. Reporte 2 sin violaciones.

8. Repositorio

<https://github.com/alejandrotapia20/CodeInspection>. Los dos reportes están en la carpeta reportes.

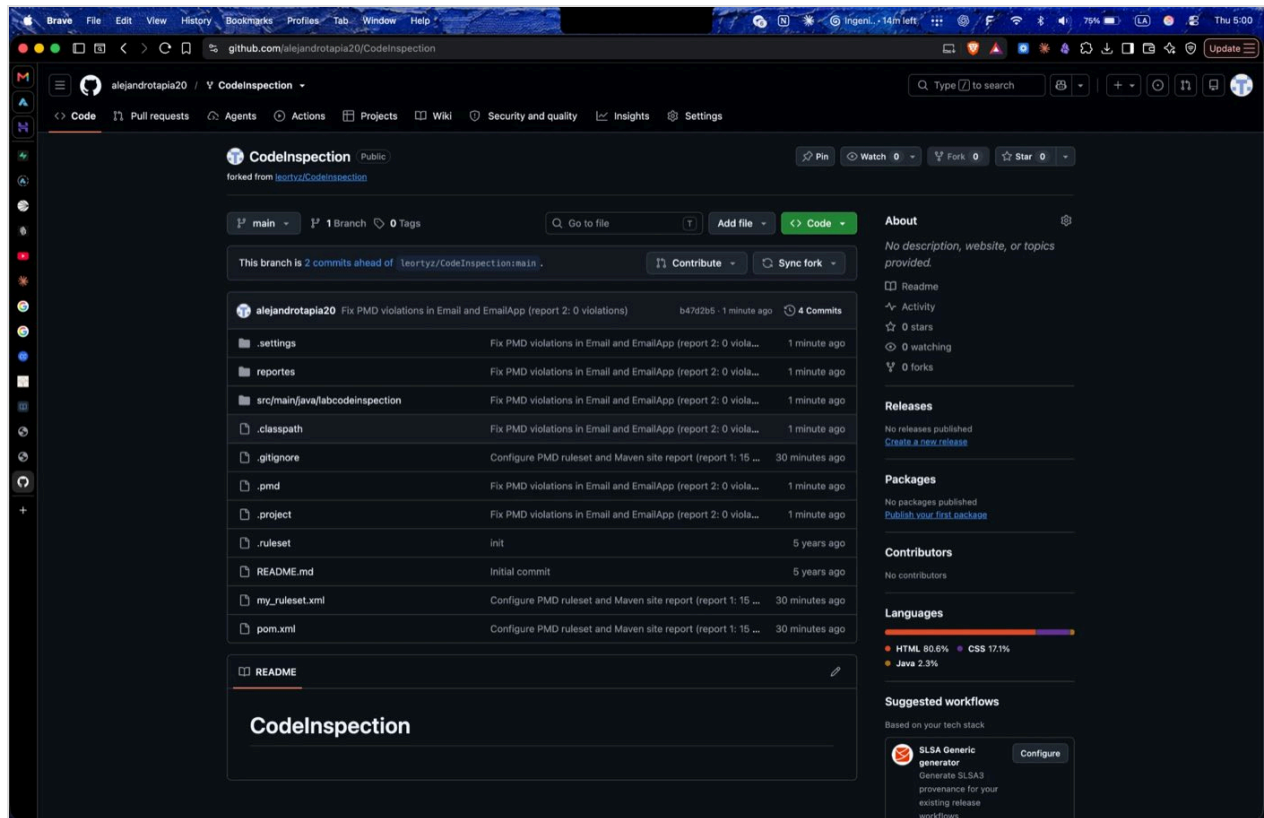


Figura 17. Repositorio con los cambios.

9. Conclusión

PMD encontró problemas que no impiden compilar pero sí afectan el mantenimiento, como atributos que debían ser `final` o conversiones de texto que dependen del idioma del sistema. También noté que Eclipse y Maven no dan exactamente el mismo resultado porque usan versiones distintas de PMD.